

(A) ALKANE

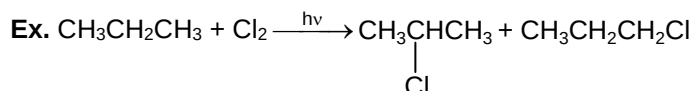
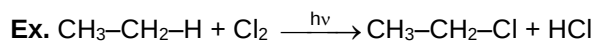
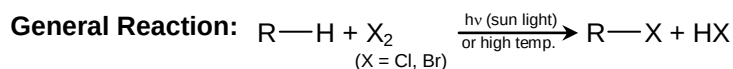
Preparation of alkanes (6-Methods)

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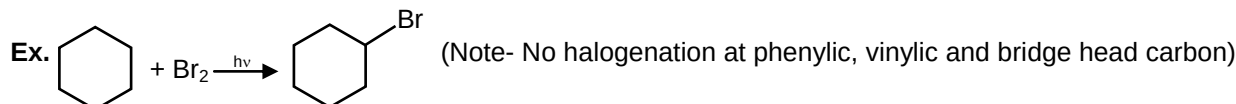
Chemical reactions of alkanes (2-Reactions)

1. Photochemical halogenation:



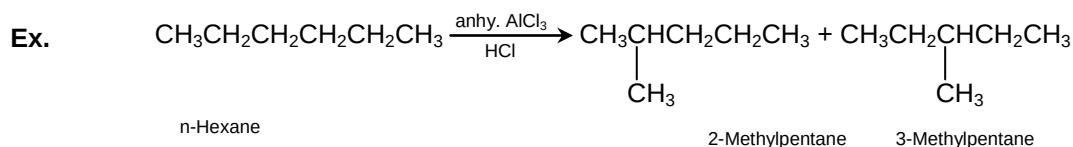
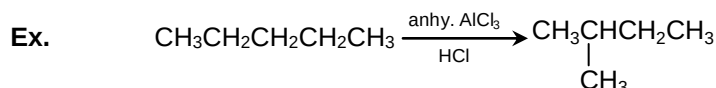
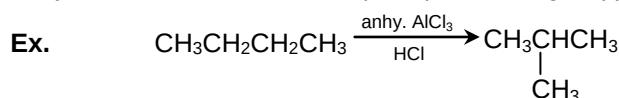
Reactivity of $\text{X}_2 = \text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$

Reactivity of H = $3^\circ\text{H} > 2^\circ\text{H} > 1^\circ\text{H}$



2. Isomerisation of alkanes:

Isomerisation is a process by which straight chain alkanes converted into branched alkanes in presence of anhydrous aluminum chloride (AlCl₃) and HCl gas (quaternary carbon can not be generated by this method).



(B) ALKENE

The alkenes are unsaturated hydrocarbons that contain one double bond (C=C). They have the general formula C_nH_{2n}. Alkenes are also known as **olefins**.

Preparation of alkenes (4-Methods)

1.	Partial reduction of alkyne: (a) Lindlar's catalyst : [H ₂ /Pd, CaCO ₃ , quinoline] (b) Rosenmund catalyst : [H ₂ /Pd, BaSO ₄ , quinoline] (c) Birch reduction [Na or Li/NH ₃ (liq.)] (Birch reduction is not observed on terminal alkynes)	$\text{R}-\text{C}\equiv\text{C}-\text{R} \xrightarrow[\text{H}_2, \text{Pd, BaSO}_4, \text{S}]{\text{H}_2, \text{Lindlar's catalyst or}} \begin{array}{c} \text{R} \quad \text{R} \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{H} \quad \text{H} \end{array}$ (cis alkene) $\text{R}-\text{C}\equiv\text{C}-\text{R} \xrightarrow{\text{Na/NH}_3(\ell)} \begin{array}{c} \text{R} \quad \text{H} \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{H} \quad \text{R} \end{array}$ (Trans alkene)
2.	Dehydrohalogenation of alkyl halides: Dehydrohalogenation is the elimination of a hydrogen and a halogen from an alkyl halide to form an alkene. [both H and X remove from adjacent positions] Rate of dehydrohalogenation of alkyl halides : $3^\circ > 2^\circ > 1^\circ$	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{R}-\text{C}-\text{C}-\text{H} \\ \quad \\ \text{H} \quad \text{X} \end{array} + \text{KOH} \xrightarrow[\Delta]{\text{alcohol}} \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{R}-\text{C}=\text{C}-\text{H} \end{array} + \text{KX} + \text{H}_2\text{O}$ More stable alkene is the major product generally. It is also known as Saytzeff alkene. (Trans alkene is always preferred if possible)



3.	Dehydration of alcohols: An alcohol is converted into an alkene by dehydration : (<i>elimination of a water molecule</i>). It is generally carried out in either of two ways: (a) By heating the alcohol with conc. sulfuric acid. (b) By passing the alcohol vapour over a catalyst Al_2O_3 (alumina) at high temperature. Rate of dehydration of alcohols: $3^\circ > 2^\circ > 1^\circ$	$ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{R}-\text{C}-\text{C}-\text{R} \\ \quad \\ \text{H} \quad \text{OH} \end{array} \xrightarrow[\Delta]{\text{acid}} \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{R}-\text{C}=\text{C}-\text{R} \\ \text{Alkene} \end{array} + \text{H}_2\text{O} $ Alcohol More stable alkene is the major product generally.
4.	Dehalogenation of vicinal dihalides: Dehalogenation of vicinal dihalides can be carried out by zinc in presence of acetic acid.	$ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{R}-\text{C}-\text{C}-\text{R} \\ \quad \\ \text{X} \quad \text{X} \end{array} + \text{Zn} \xrightarrow{\Delta} \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{R}-\text{C}=\text{C}-\text{R} \end{array} + \text{ZnX}_2 $

Chemical reactions of alkenes [6-Reactions] :

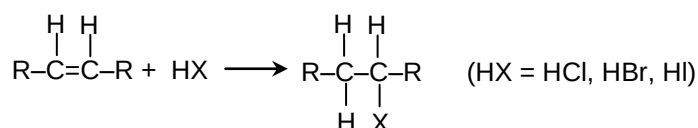
Markovnikov's rule : The rule states that the negative part of the attacking species add on the carbon atom containing less number of hydrogen atom and positive part add on the carbon atom containing more number of hydrogen atoms.

1. Addition of hydrogen halides (Hydrohalogenation)

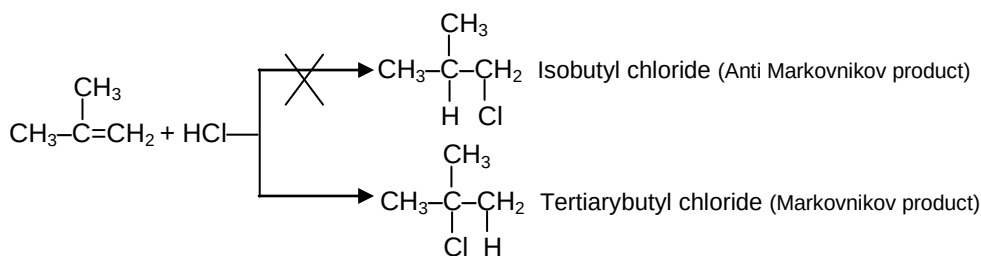
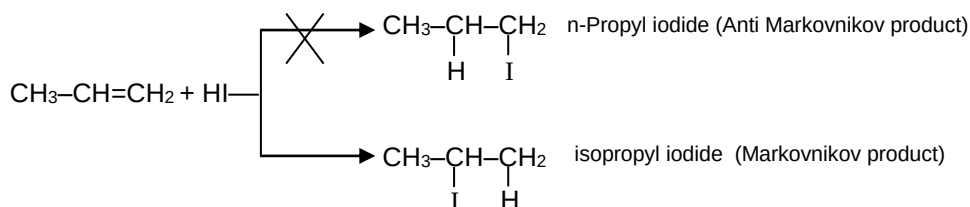
An alkene reacts with conc. hydrogen chloride, hydrogen bromide, or hydrogen iodide to give the corresponding alkyl halide.

Major Product : According to Markovnikov's rule.

General Reaction:



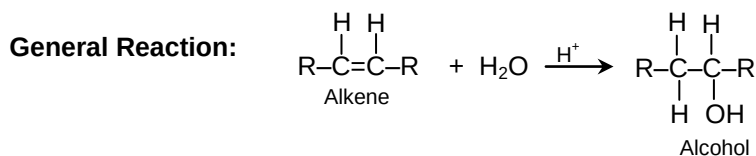
Examples:



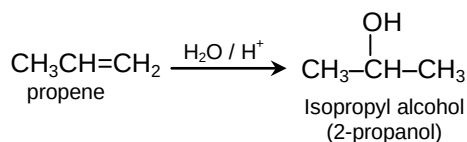


2. Addition of water (Hydration)

Alkenes add water in the presence of an acid to give alcohols. The addition takes place according to **Markovnikov rule**.



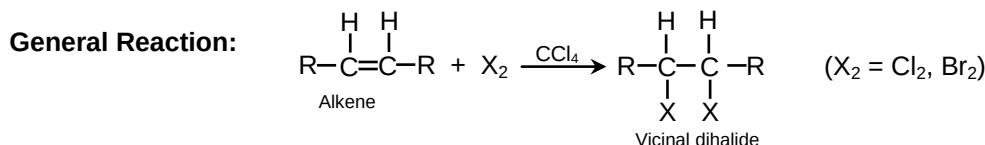
Example:



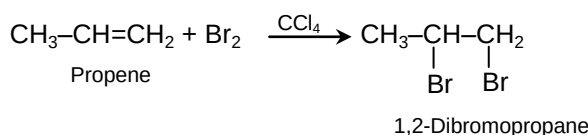
3. Addition of halogens (Halogenation)

Halogen add to alkenes to form vicinal dihalides. This reaction is the best method for preparing vicinal dihalides. The reaction is carried out in an inert solvent like carbon tetrachloride.

Reaction with F_2 is very tremendous while with I_2 it is very slow and reversible.



Example:

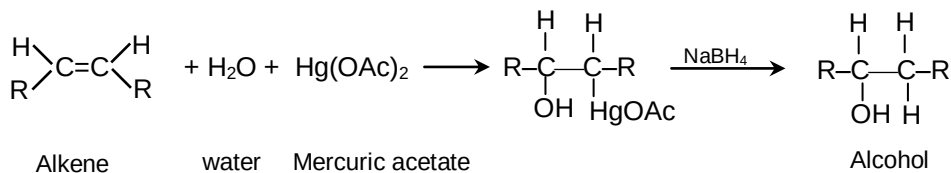


4. Hydration by $\text{Hg}(\text{OAc})_2$, $\text{H}_2\text{O}/\text{NaBH}_4$ (Oxymercuration-demercuration)

Alkenes react with mercuric acetate in the presence of water to give hydroxyl-mercurial compounds which on reduction yield alcohols.

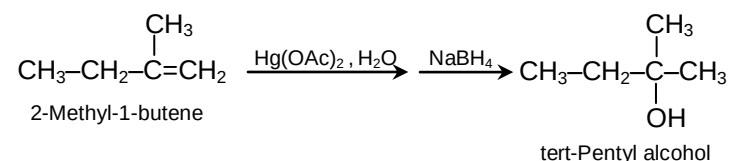
- Oxymercuration-demercuration process gives alcohols corresponding to **Markovnikov** addition of water to the carbon-carbon double bond.

General Reaction:



Acetate = $(-\text{OAc}) = (\text{CH}_3\text{COO}-)$

Example.



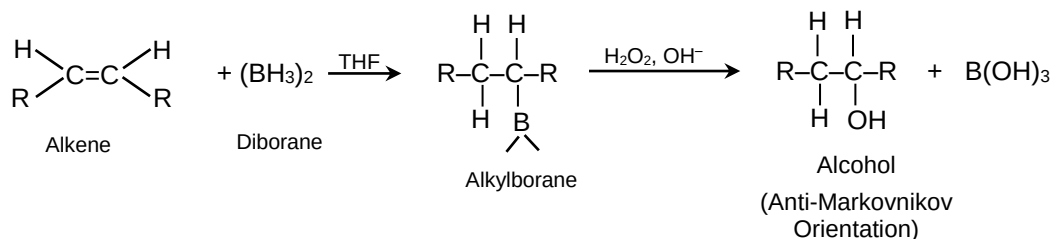


5. Hydration by $B_2H_6/H_2O_2, OH^-$ (Hydroboration-oxidation)

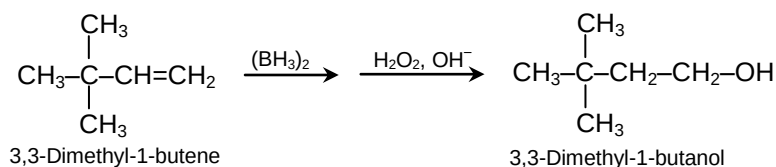
With the reagent diborane (B_2H_6), alkenes undergo hydroboration to give trialkylboranes (R_3B) which on oxidation give alcohols.

- The hydroboration–oxidation process gives alcohols corresponding to **anti-Markovnikov** addition of water to the carbon-carbon double bond.
- It is syn addition of H and OH.
- Ethers like tetrahydrofuran (THF) acts as inert solvent with diborane.

General Reaction:



Example:

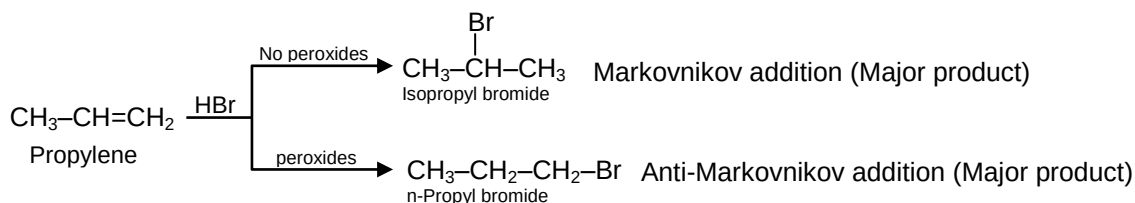


6. Peroxide effect (Addition of HBr)

In the absence of peroxides, hydrogen bromide adds to alkenes according to Markovnikov's rule but in the presence of peroxides, the position of addition is exactly reversed. This reversal of the orientation of addition caused by the presence of peroxides is known as the **peroxide effect/Khrasch effect**.

Only the addition of hydrogen bromide shows the peroxide effect. The presence or absence of peroxides has no effect on the orientation of addition of hydrogen chloride, hydrogen iodide, sulfuric acid, water, etc.

Example:





(C) ALKYNE

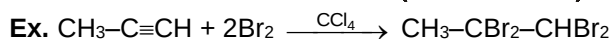
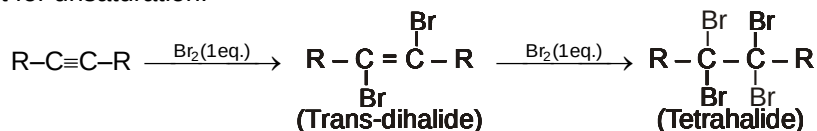
The alkynes are unsaturated hydrocarbons that contain one triple bond ($C\equiv C$). They have the general formula C_nH_{2n-2} and the triple bond is known as the 'acetylenic bond'. Many alkynes have been found in nature.

Preparation of alkynes (4-Methods)		
1.	Hydrolysis of carbides	1. $CaC_2 + 2HOH \rightarrow CH\equiv CH + Ca(OH)_2$; 2. $Mg_2C_3 + 4HOH \rightarrow CH_3-C\equiv CH + 2Mg(OH)_2$
2.	Dehydrohalogenation of gem and vicinal dihalides. Dihalides in which two halogen atoms are attached to two adjacent carbon atoms are known as vicinal dihalides. Dihalides in which two halogen atoms are attached to same carbon atom are known as geminal dihalides.	$ \begin{array}{c} H \quad H \\ \quad \\ R-C-C-R \\ \quad \\ Br \quad Br \\ \text{vic-dibromide} \end{array} \xrightarrow[\Delta]{2NaNH_2} R-C\equiv C-R + 2NaBr $ $ \begin{array}{c} H \quad H \\ \quad \\ CH_3-C-C-CH_3 \\ \quad \\ Br \quad Br \end{array} \xrightarrow[\Delta]{2NaNH_2} CH_3-C\equiv C-CH_3 $ $ \begin{array}{c} CH_3-CH-CH_2 \\ \quad \\ Br \quad Br \end{array} \xrightarrow[\Delta]{2NaNH_2} CH_3-C\equiv C-H $
3.	Dehalogenation of tetrahaloalkanes	$ \begin{array}{c} X \quad X \\ \quad \\ R-C-C-R' \\ \quad \\ X \quad X \end{array} \xrightarrow[\Delta]{Zn} R-C\equiv C-R' + 2ZnX_2 $ $ CH_3-CBr_2-CHBr_2 \xrightarrow[\Delta]{Zn} CH_3-C\equiv CH + 2ZnBr_2 $
4.	1,1,1-trihaloalkane with Ag (Silver) powder	$2R-CX_3 + 6Ag \longrightarrow R-C\equiv C-R + 6AgX$ $2CHI_3 + 6Ag \longrightarrow H-C\equiv C-H + 6AgI$ $2CH_3-CCl_3 + 6Ag \longrightarrow CH_3-C\equiv C-CH_3$

Chemical reactions of alkynes (4-Reactions)

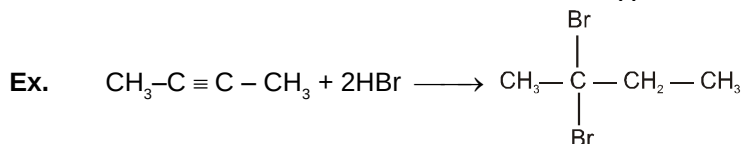
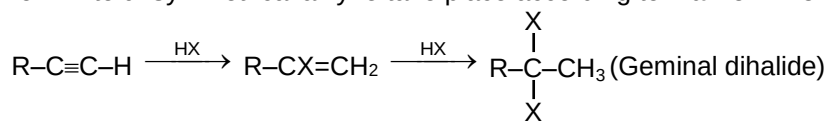
(1) Addition of Halogens :

Halogens like bromine or chlorine add up to alkyne to form trans dihalides and further addition of halogen give tetrahalo alkane. It is an example of electrophilic addition reaction. This reaction is used as a test for unsaturation.



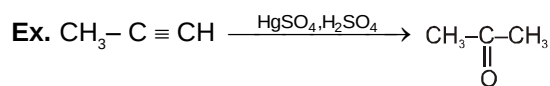
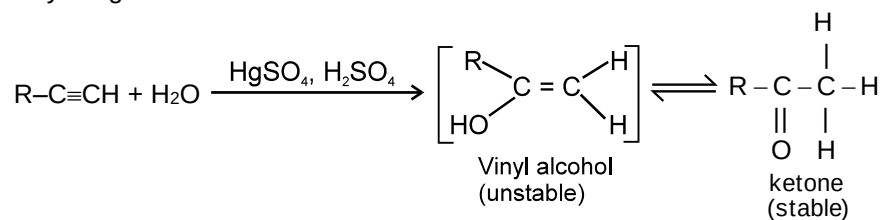
(2) Addition of Hydrogen halides :

Addition of HX to unsymmetrical alkyne take place according to **Markovnikov's rule**.

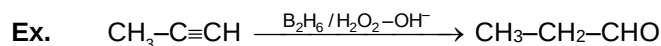
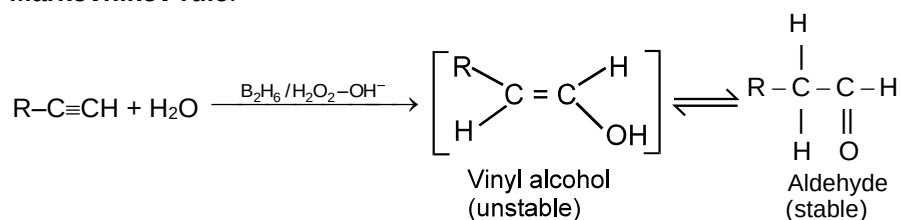
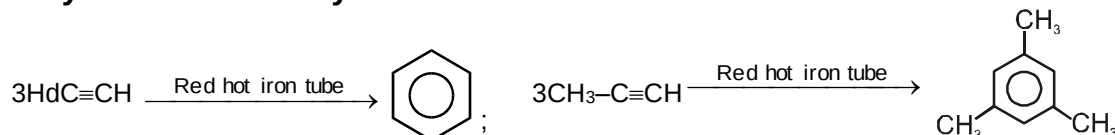


**(3) Addition of water :**

- (A) Alkyne react with water to form carbonyl compounds, in accordance with the **Markovnikov rule**.
Catalyst- Hg^{2+}



- (B) Alkynes react with $\text{B}_2\text{H}_6/\text{H}_2\text{O}_2, \text{OH}^-$ to form carbonyl compounds, in accordance with the **anti-Markovnikov rule**.

**4. Polymerization of alkyne :****Lab Test :****1. Test for unsaturation ($>\text{C}=\text{C}<$, $-\text{C}\equiv\text{C}-$) :**

Unsaturated compound give bromine water test ($\text{Br}_2 + \text{H}_2\text{O}$) and Baeyer's test (Cold dil. alkaline KMnO_4).

Functional Group	Reagent	Observation
$>\text{C}=\text{C}<$, $-\text{C}\equiv\text{C}-$	Bromine water test ($\text{Br}_2 + \text{H}_2\text{O}$)	Red-brown colour disappears
$>\text{C}=\text{C}<$, $-\text{C}\equiv\text{C}-$	Baeyer's reagent (Cold, dil. alk. KMnO_4)	Purple colour disappear

Note : (i) Benzene does not give bromine water test and Baeyer's test.
(ii) phenol and aniline both give bromine water test but not Baeyer's test.

2. Test for terminal alkyne [$\text{R}-\text{C}\equiv\text{C}-\text{H}$]

Terminal alkyne give Tollen's test and ammonical cuprous chloride test

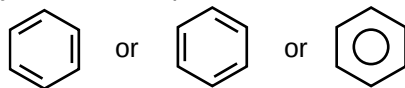
Functional Group	Reagent	Observation
$[\text{R}-\text{C}\equiv\text{C}-\text{H}]$	Tollen's reagent [$\text{AgNO}_3 + \text{NH}_4\text{OH}$]	white precipitate
$[\text{R}-\text{C}\equiv\text{C}-\text{H}]$	Ammonical cuprous chloride ($\text{Cu}_2\text{Cl}_2 + \text{NH}_4\text{OH}$)	Red precipitate



(D) BENZENE

Benzene is an aromatic compound with alternate π -bonds having 6 delocalized π -electrons. It is a colourless & highly flammable liquid.

It is represented as

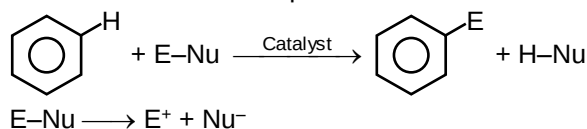


Preparation of benzene (3-Methods)

1.	From alkyne When acetylene is passed through a red hot metallic tube, cyclic polymerization takes place and benzene is formed	$ \begin{array}{l} \text{1. } \begin{array}{c} \text{HC} \\ \\ \text{HC} \end{array} + \begin{array}{c} \text{HC} \\ \equiv \\ \text{CH} \end{array} \xrightarrow[1500-2000^\circ\text{C}]{\text{Red hot tube}} \text{Benzene} \\ \text{2. } 3\text{CH}_3\text{-C}\equiv\text{CH} \xrightarrow{\text{Red hot tube}} \text{Mesitylene} \end{array} $
2.	From phenol Benzene is obtained by distillation of phenol with zinc dust.	$ \begin{array}{l} \text{1. } \text{Phenol} + \text{Zn} \xrightarrow{\text{Distillation}} \text{Benzene} + \text{ZnO} \\ \text{(-OH group of the benzene nucleus is replaced by -H in this reaction)} \\ \text{2. } \text{p-Cresol} + \text{Zn} \xrightarrow{\text{Distillation}} \text{p-Toluene} + \text{ZnO} \end{array} $
3.	From decarboxylation of benzoic acid Benzene is conveniently prepared in the laboratory by heating the mixture of benzoic acid/sodium benzoate with soda lime (NaOH + CaO). This reaction is called sodalime decarboxylation.	$ \begin{array}{l} \text{1. } \text{Sodium benzoate} + \text{NaOH} \xrightarrow[\text{Heat}]{\text{CaO}} \text{Benzene} + \text{Na}_2\text{CO}_3 \\ \text{2. } \text{p-Toluic acid} + \text{NaOH} \xrightarrow[\text{Heat}]{\text{CaO}} \text{p-Toluene} + \text{Na}_2\text{CO}_3 \end{array} $

Chemical reactions of benzene [6-Reactions]

The reactions in which hydrogen atom of the benzene ring is replaced by an electrophile (electron deficient specie) are called electrophilic aromatic substitution reactions. Benzene undergoes electrophilic substitution reaction because it is an electron rich system due to delocalized π -electrons. The reactions can be represented as :



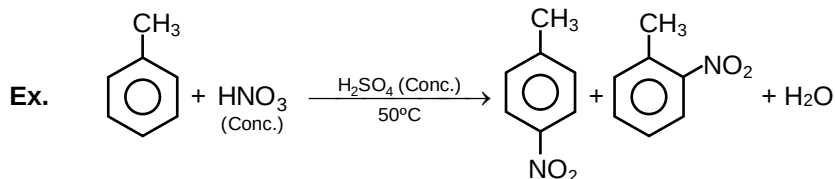
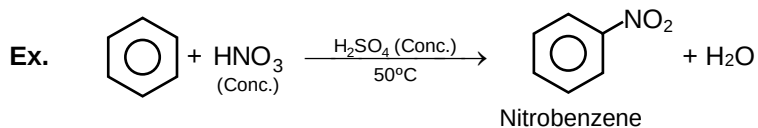


Electrophilic substitution reactions of Benzene

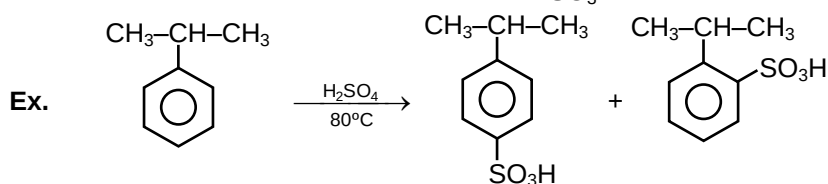
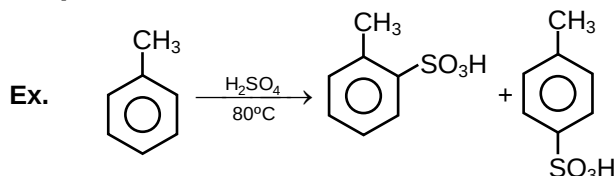
	Substrate	Reagent	Electrophile	Product	Name of reaction
1.		Conc. HNO_3 / Conc. H_2SO_4 Nitrating mixture	NO_2^+		Nitration of benzene
2.		Conc. H_2SO_4 + SO_3	SO_3		Sulphonation of benzene
3.		Cl_2 / AlCl_3	Cl^+		Chlorination of benzene
4.		Br_2 / AlCl_3	Br^+		Bromination of benzene
5.		CH_3Cl / AlCl_3	CH_3^+		Friedal Craft's alkylation of benzene
6.		$\text{CH}_3\text{C}(=\text{O})\text{Cl}$ / AlCl_3	$\text{CH}_3-\text{C}^+=\text{O}$		Friedal Craft's acylation of benzene

Note- In the halogenation and Friedal Craft's reactions, the catalyst involved can be either of AlCl_3 , FeCl_3 , SnCl_4 , BF_3 , BCl_3 or ZnCl_2 (Lewis acid). However, AlCl_3 provides satisfactory results.

(1) Nitration

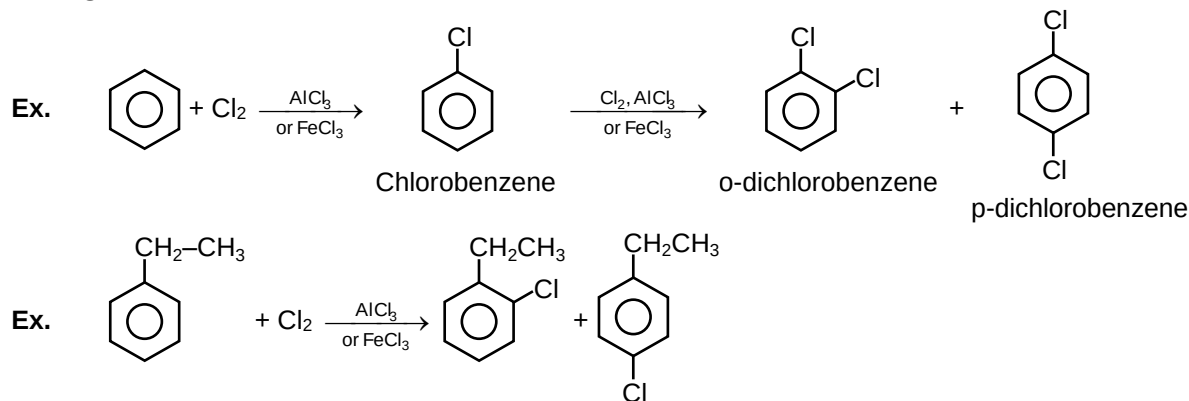


(2) Sulphonation



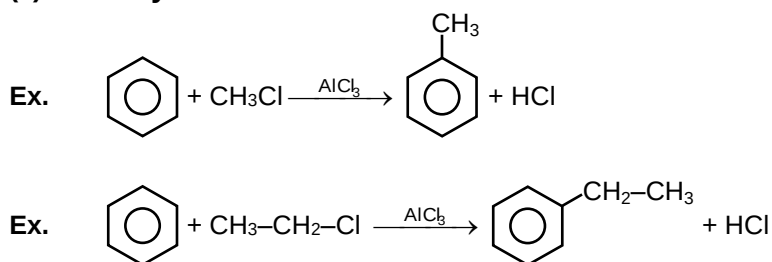


(3) Halogenation

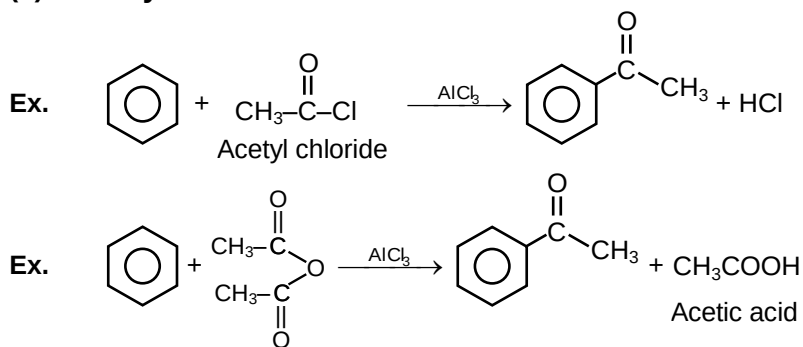


(4) Friedel-Craft's reaction

(a) Alkylation



(b) Acylation



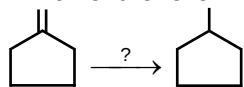


Exercise

ONLY ONE OPTION CORRECT TYPE

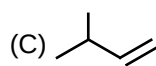
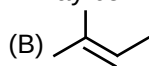
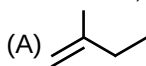
PART-A (Alkane)

1. Which of the following catalyst can be used for the given conversion ?



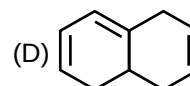
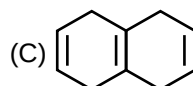
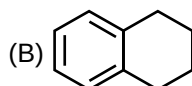
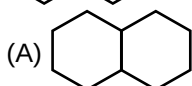
- (A) H_2/Ni (B) H_2/Pd (C) H_2/Pt (D) All of these

2. $\text{X} \xrightarrow{\text{H}_2/\text{Ni}}$; X may be



- (D) All of these

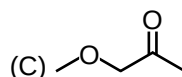
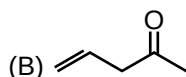
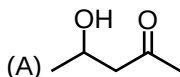
3. $\xrightarrow[\text{room temp.}]{\text{H}_2/\text{Ni}}$ Product



4. $\xrightarrow{\text{Reagent}}$, Reagent is

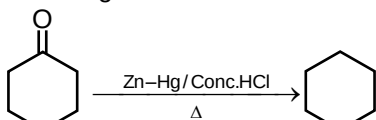
- (A) H_2/Pd (B) LiAlH_4 (C) $\text{Zn-Hg}/\text{conc. HCl}$ (D) All of these

5. In which case Clemensen reduction should be avoided.



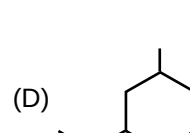
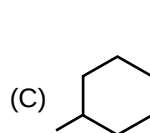
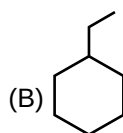
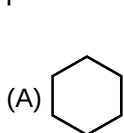
- (D) All of these

6. For the given conversion A should not be

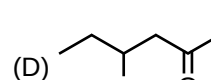
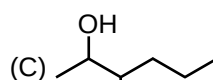
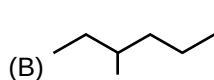
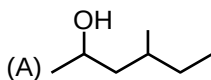


- (A) Halogen (B) Alcohol (C) Carboxylic acid (D) All of these

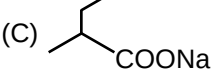
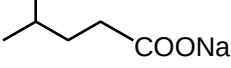
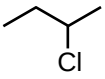
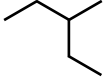
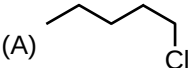
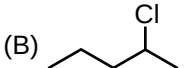
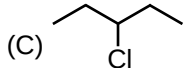
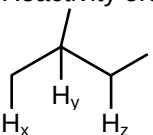
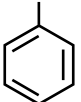
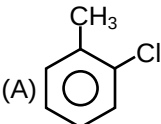
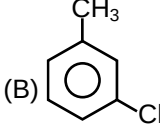
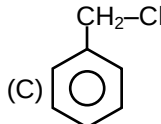
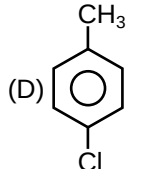
7. $\xrightarrow{\text{NH}_2\text{-NH}_2/\text{KOH}}$ P, Product P is

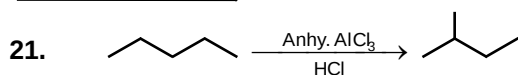


8. $\xrightarrow{\text{NH}_2\text{-NH}_2/\text{KOH}}$ P
Product P is





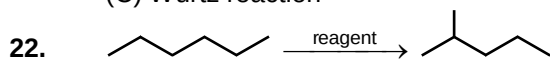
9. $\text{CH}_3\text{COONa} \xrightarrow[\Delta]{\text{reagent}} \text{CH}_4$; reagent is
 (A) $\text{NH}_2\text{-NH}_2 / \text{KOH}$ (B) $\text{Zn-Hg} / \text{HCl}$ (C) $\text{NaOH} + \text{CaO}$ (D) All of these
10. $\text{A} \xrightarrow[\Delta]{\text{NaOH} + \text{CaO}} \text{CH}_3\text{-CH}_2\text{-CH}_3$; A can be
 (A) $\text{CH}_3\text{CH}_2\text{COONa}$ (B) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COONa}$
 (C)  (D) $\text{CH}_3\text{CH}_2\text{CH}_2\text{ONa}$
11.  $\xrightarrow[\Delta]{\text{NaOH} + \text{CaO}}$ P; Product P is
 (A) Butane (B) 2-Methylbutane (C) 3-Methylbutane (D) Pentane
12. $\text{CH}_3\text{-CH}_2\text{-Cl} \xrightarrow{\text{Na / Dry ether}}$ P; Product P is
 (A) Ethane (B) Propane (C) Butane (D) Pentane
13. $\text{A} \xrightarrow{\text{Na / Dry ether}}$ ; A may be
 (A) Chloromethane (B) Chloroethane (C) 1-Chloropropane (D) 2-Chloropropane
14. Preparation of alkane by Wurtz reaction the reactant used is
 (A) Alkyl halide (B) Acid halide (C) Both A & B (D) None of these
15. $\text{CH}_3\text{-CH}_2\text{-Cl} \xrightarrow{2\text{Li}} \text{X} \xrightarrow{\text{CuI}} \text{Y} \xrightarrow{\text{CH}_3\text{-Cl}} \text{Z}$; Final product Z is
 (A) Ethane (B) Propane (C) Butane (D) Pentane
16.  $\xrightarrow{2\text{Li}} \text{X} \xrightarrow{\text{CuI}} \text{Y} \xrightarrow{\text{Reagent}}$ , reagent is
 (A) $\text{CH}_3\text{-Cl}$ (B) $\text{CH}_3\text{-CH}_2\text{-Cl}$ (C) $\text{CH}_3\text{-CH(Cl)-CH}_3$ (D) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-Cl}$
17. $\text{CH}_4 + \text{X}_2 \xrightarrow[\text{or high temp}]{h\nu} \text{CH}_3\text{-X}$
 Order of reactivity of halogen is
 (A) $\text{I}_2 > \text{Br}_2 > \text{Cl}_2 > \text{F}_2$ (B) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$ (C) $\text{Br}_2 > \text{Cl}_2 > \text{I}_2 > \text{F}_2$ (D) $\text{Cl}_2 > \text{Br}_2 > \text{F}_2 > \text{I}_2$
18.  $\xrightarrow{\text{Cl}_2/h\nu}$ P
 Product may be
 (A)  (B)  (C)  (D) All of these
19. Reactivity order of Hydrogen for the given reaction is
 + $\text{Br}_2 \xrightarrow{h\nu}$
 (A) $\text{X} > \text{Y} > \text{Z}$ (B) $\text{Z} > \text{X} > \text{Y}$ (C) $\text{Y} > \text{Z} > \text{X}$ (D) $\text{Z} > \text{Y} > \text{X}$
20.  $\xrightarrow{\text{Cl}_2/h\nu}$ P
 P may be
 (A)  (B)  (C)  (D) 



This reaction is known as

- (A) Isomerisation of alkane
(C) Wurtz reaction

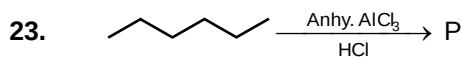
- (B) Polymerisation of alkane
(D) None of these



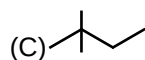
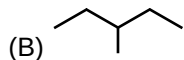
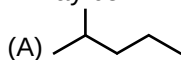
reagent may be

- (A) $\text{NH}_2\text{-NH}_2$ / KOH (B) Zn-Hg / HCl

- (C) Anhy. AlCl_3 / HCl (D) Na / Dry ether

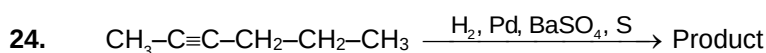


P may be



- (D) A & B both

PART-B (Alkene)

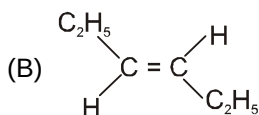
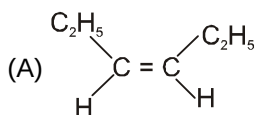
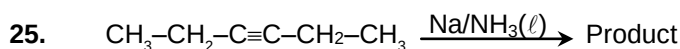


- (A) $\text{CH}_3\text{-CH=CH-CH}_3$

- (B) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-CH}_3$

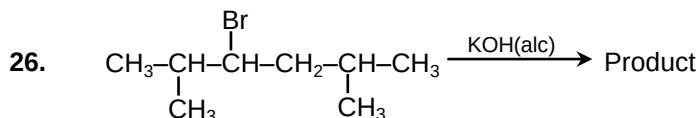
- (C) $\text{CH}_3\text{-CH=CH-CH}_2\text{-CH}_2\text{-CH}_3$ (cis)

- (D) $\text{CH}_3\text{-CH=CH-CH}_2\text{-CH}_2\text{-CH}_3$ (trans)

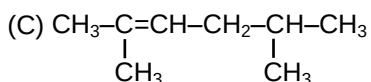
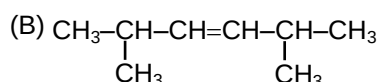
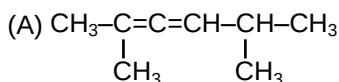


- (C) Both A & B

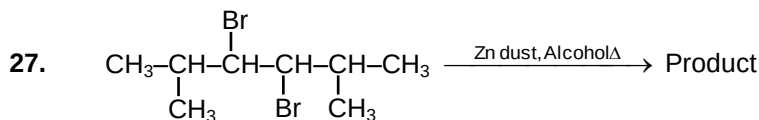
- (D) None of these



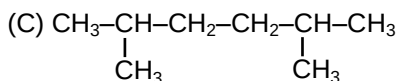
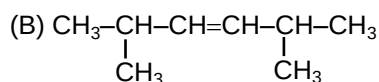
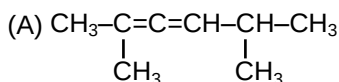
Product is:



- (D) None of these



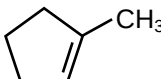
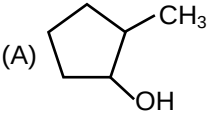
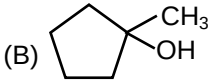
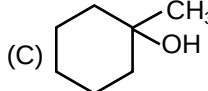
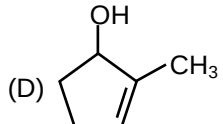
Product is:



- (D) None of these

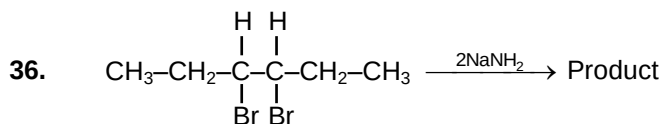




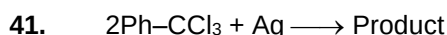
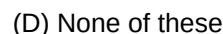
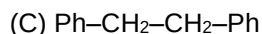
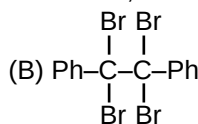
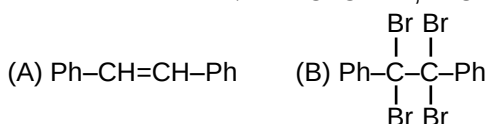
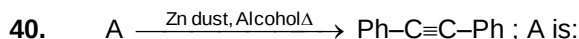
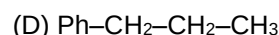
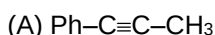
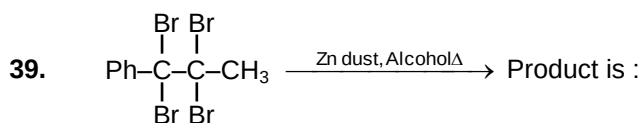
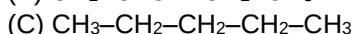
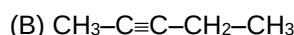
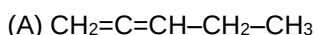
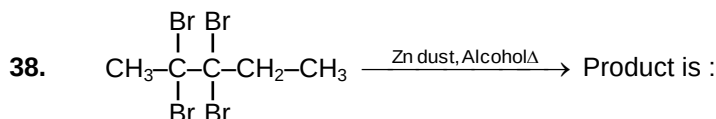
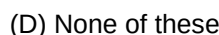
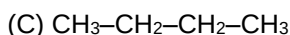
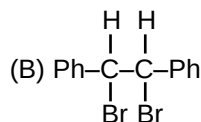
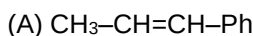
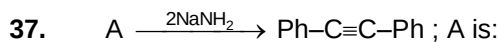
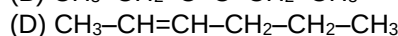
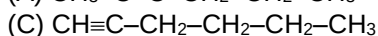
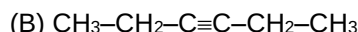
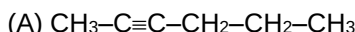
28. $\text{Ph}-\text{CH}=\text{CH}_2 + \text{HBr} \longrightarrow \text{Product}$
 (A) $\text{Ph}-\text{CH}_2-\underset{\text{Br}}{\text{CH}_2}$ (B) $\text{Ph}-\underset{\text{Br}}{\text{CH}}=\text{CH}$ (C) $\text{Ph}-\underset{\text{Br}}{\text{CH}}-\text{CH}_3$ (D) $\text{Ph}-\text{CH}_2-\text{CHBr}_2$
29. $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_3 \xrightarrow{\text{H}_2\text{O}/\text{H}^+} \text{Product}$
 (A) $\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\text{CH}_2-\text{CH}_3$ (B) $\text{CH}_3\text{CH}=\text{CH}-\text{CH}_2-\text{OH}$
 (C) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{OH}$ (D) None of these
30. $\text{CH}_3-\text{CH}=\text{CH}_2 + \text{Br}_2 \xrightarrow{\text{CCl}_4} \text{Product}$
 (A) $\text{CH}_2-\text{CH}_2-\underset{\text{Br}}{\text{CH}_2}$ (B) $\text{CH}_3-\underset{\text{Br}}{\text{CH}}-\text{CH}_3$
 (C) $\text{CH}_3-\underset{\text{Br}}{\text{CH}}-\underset{\text{Br}}{\text{CH}_2}$ (D) None of these
31.  $\xrightarrow[\text{NaBH}_4]{\text{Hg}(\text{OAc})_2, \text{H}_2\text{O}}$ Product
 (A)  (B)  (C)  (D) 
32. $\text{CH}_3-\text{CH}_2-\text{CH}=\text{CH}_2 \xrightarrow{\text{A}} \text{CH}_3-\text{CH}_2-\underset{\text{OH}}{\text{CH}}-\text{CH}_3$
 Reagent A will be ?
 (A) $\text{Hg}(\text{OA})_2, \text{H}_2\text{O} / \text{NaBH}_4$ (B) $\text{B}_2\text{H}_6, \text{H}_2\text{O}_2 + \text{OH}^-$
 (C) Both (A) and (B) (D) None of these
33. $\text{CH}_3-\text{CH}_2-\text{CH}=\text{CH}_2 \xrightarrow{\text{A}} \text{CH}_3-\text{CH}_2-\text{CH}_2-\underset{\text{OH}}{\text{CH}_2}$
 Reagent A will be ?
 (A) $\text{Hg}(\text{OA})_2, \text{H}_2\text{O} / \text{NaBH}_4$ (B) $\text{B}_2\text{H}_6, \text{H}_2\text{O}_2 + \text{OH}^-$
 (C) Both (A) and (B) (D) None of these
34. $\text{Ph}-\text{CH}=\text{CH}_2 + \text{HBr} \xrightarrow{\text{peroxide}} \text{Product}$
 (A) $\text{Ph}-\underset{\text{Br}}{\text{CH}}-\underset{\text{Br}}{\text{CH}_2}$ (B) $\text{Ph}-\underset{\text{Br}}{\text{CH}}=\text{CH}$ (C) $\text{Ph}-\text{CHBr}-\text{CH}_3$ (D) $\text{Ph}-\text{CH}_2-\text{CH}_2\text{Br}$
35. $\text{Ph}-\text{CH}=\text{CH}_2 + \text{HCl} \xrightarrow{\text{peroxide}} \text{Product}$
 (A) $\text{Ph}-\underset{\text{Cl}}{\text{CH}_2}-\underset{\text{Cl}}{\text{CH}_2}$ (B) $\text{Ph}-\underset{\text{Cl}}{\text{CH}}=\text{CH}$ (C) $\text{Ph}-\underset{\text{Cl}}{\text{CH}}-\text{CH}_3$ (D) $\text{Ph}-\text{CH}_2-\text{CHCl}_2$



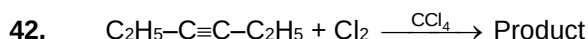
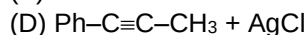
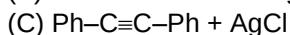
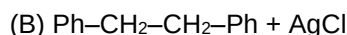
PART-C (Alkyne)



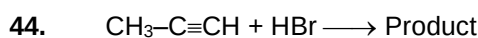
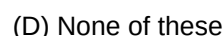
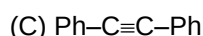
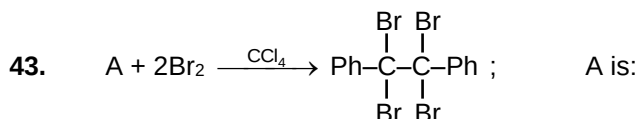
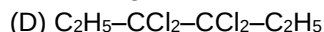
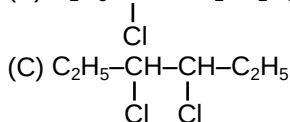
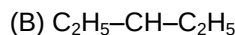
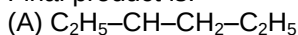
Product is:



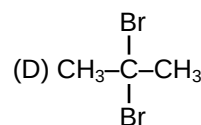
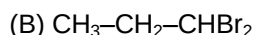
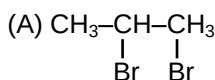
Product is:



Final product is:



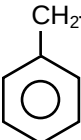
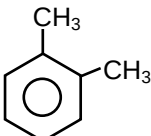
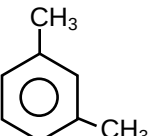
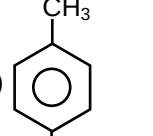
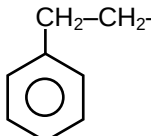
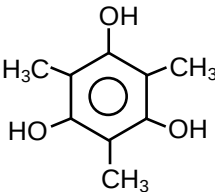
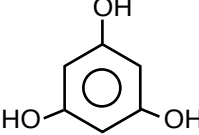
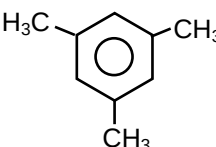
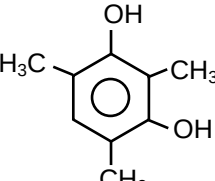
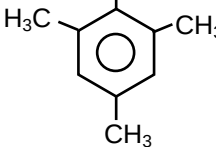
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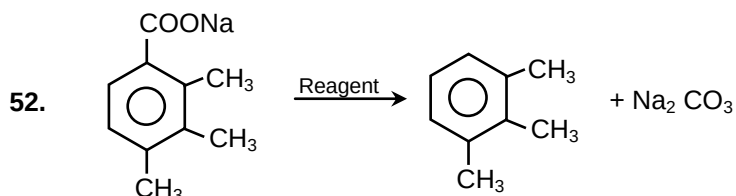


45. Alkyne react with water to form _____ compounds.
 (A) carboxylic (B) ester (C) carbonyl (D) Anhydride
46. $\text{CH}_3\text{-C}\equiv\text{CH} \xrightarrow[\text{H}_2\text{O}_2/\text{OH}^-]{\text{B}_2\text{H}_6}$ Product
 Product is:
 (A) $\text{CH}_3\text{-}\underset{\text{OH}}{\text{C}}\text{-CH}_3$ (B) $\text{CH}_3\text{CH}_2\text{-CHO}$ (C) $\text{CH}_3\text{-}\overset{\text{O}}{\underset{\text{O}}{\text{C}}}\text{-CH}_3$ (D) $\text{CH}_3\text{-CH}_2\text{-}\underset{\text{OH}}{\text{CH}_2}$
47. $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CCH}_2\text{CH}_3 + \text{H}_2\text{O} \xrightarrow{\text{HgSO}_4}$ Product
 Product is:
 (A) $\text{CH}_3\text{CH}_2\overset{\text{O}}{\underset{\text{O}}{\text{C}}}\text{CH}_2\text{CH}_2\text{CH}_3$ (B) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
 (C) both A & B (D) None of these

PART-D (Benzene)

48. $x \text{ HC}\equiv\text{CH} + y \text{ CH}_3\text{-CH}_2\text{-C}\equiv\text{CH} \longrightarrow$  ; Find the value of 'x' and 'y' ?
 (A) x = 1, y = 2 (B) x = 2, y = 1 (C) x = 2, y = 2 (D) x = 1, y = 1
49. Which of the following will not form in the following reaction ?
 $2\text{CH}_3\text{-C}\equiv\text{C-H} + \text{HC}\equiv\text{CH} \xrightarrow[\text{tube}]{\text{Red hot}} ?$
 (A)  (B)  (C)  (D) 
50. Phenol upon distillation with zinc dust leads to the formation of
 (A) Toluene (B) Benzene (C) Cyclohexane (D) n-hexane
51.  $\xrightarrow[\text{distillation}]{\text{Zn powder}} ?$
 (A)  (B) 
 (C)  (D) 





Reagent may be -

- (A) Soda lime (B) Zinc dust (C) Red hot tube (D) none of these

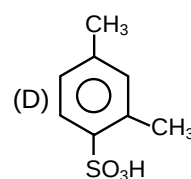
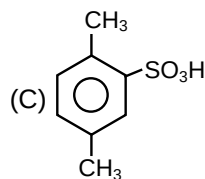
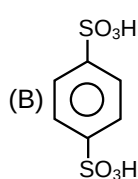
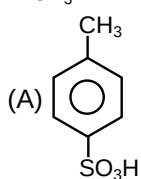
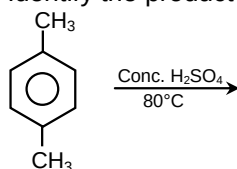
53. Benzene upon addition with the mixture of conc. HNO_3 and conc. H_2SO_4 undergoes

- (A) nitration (B) sulphonation
(C) both nitration and sulphonation (D) neither nitration nor sulphonation

54. Benzene formswith hot concentrated sulphuric acid (80°C temperature)

- (A) Benzene sulphonic acid (B) Benzoic acid
(C) Both (D) None of the above

55. Identify the product :



Answers

- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (D) | 2. (D) | 3. (B) | 4. (C) | 5. (D) |
| 6. (B) | 7. (B) | 8. (C) | 9. (C) | 10. (B) |
| 11. (B) | 12. (C) | 13. (C) | 14. (A) | 15. (B) |
| 16. (B) | 17. (B) | 18. (D) | 19. (C) | 20. (C) |
| 21. (A) | 22. (C) | 23. (D) | 24. (C) | 25. (B) |
| 26. (C) | 27. (B) | 28. (C) | 29. (A) | 30. (C) |
| 31. (B) | 32. (A) | 33. (B) | 34. (D) | 35. (C) |
| 36. (B) | 37. (B) | 38. (B) | 39. (A) | 40. (B) |
| 41. (C) | 42. (D) | 43. (C) | 44. (D) | 45. (C) |
| 46. (B) | 47. (A) | 48. (B) | 49. (D) | 50. (B) |
| 51. (B) | 52. (A) | 53. (A) | 54. (A) | 55. (C) |